

Revised Thesis Outline

A TSUKIMI-Centered Lunar Subsurface Thermal Pipeline
with Jacobian-Coupled RTM Inversion and Ice Survivability Mapping

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Core reframing. Your thesis is no longer 'a 1D thermal model validated against Apollo.' It is: *'A 20 m/pixel subsurface thermal pipeline for the lunar south pole, coupled to TSUKIMI's terahertz radiative transfer model via Jacobian-based joint inversion, with downstream ice survivability mapping.'* Apollo validation becomes one paragraph in Methods. The discrete 3-layer model becomes a historical comparison figure showing why continuous properties matter. Everything you built carries forward — nothing is wasted.

1. Thesis Chapter Structure

Chapter 1: Introduction

1.1 Scientific context. Lunar polar volatiles, Artemis program, the need for subsurface thermal profiles at exploration-relevant scales (20 m).

1.2 The TSUKIMI mission. Terahertz radiometry in the sensing gap between Diviner IR and Chang'E microwave. Your role: providing the thermal profile input.

1.3 Current state of thermal modeling. From Vasavada two-layer → Hayne H-parameter → Martinez-Siegler low-T correction → Burger microphysical. The progression toward physically realistic, continuous properties.

1.4 Objectives. (1) Build a pipeline producing $T(z,t)$ at 20 m/pixel, 0–3 m depth, with shadow modeling. (2) Couple to TSUKIMI RTM via Jacobian inversion. (3) Produce ice survivability maps as downstream products. (4) Quantify sensitivity of all outputs to conductivity model choice.

Chapter 2: Thermal Model Physics

2.1 Governing equation. 1D heat equation with depth- and temperature-dependent properties. Your existing Crank-Nicolson solver — proven, Numba-accelerated.

2.2 Continuous regolith properties. The H-parameter model: $\rho(z) = 1800 - 700 \cdot \exp(-z/H)$, $K(T,z) = K_c(z) \cdot [1 + \chi(T/350)^3]$. Full equations with parameter values from Hayne et al. (2017).

2.3 Low-temperature conductivity correction. Martinez-Siegler (2021) modification to K_c below 150 K. Implemented as switchable option. Physical basis in amorphous-solid phonon transport.

2.4 Why not discrete layers? Brief comparison figure: your original 3-layer model vs. continuous H-parameter at 80°S. Show the unphysical density jumps and their effect on the thermal gradient in the top 20 cm. *This is where your old work lives — as motivation for the upgrade, not as a discarded mistake.*

2.5 Computational grid. Geometric spacing: $\Delta z_0 = 2$ mm, growth factor 0.15, ~55 layers to 3 m. ≥ 10 points in diurnal skin depth. Output resampled to 5–10 cm for TSUKIMI delivery.

2.6 Boundary conditions. Surface energy balance (direct + scattered solar + terrain IR – $\epsilon \sigma T^4$). Bottom: geothermal flux $Q_b = 18$ mW/m².

2.7 Validation. Brief Apollo 15/17 sanity check (~3–6 K RMSE). Cross-check against Chang'E-4 and ChaSTE in situ measurements. Primary validation: Diviner polar bolometric temperatures (§2.8).

Chapter 3: Illumination and Shadow Modeling

3.1 LOLA DEM acquisition and preprocessing. 20 m south pole DEM from PGDA. Nested-grid horizon computation (20 m near / 80 m mid / 240 m far).

3.2 Horizon-based direct illumination. Your existing ray-tracer upgraded to 20 m. 720 azimuthal directions. SPICE ephemeris for proper obliquity treatment.

3.3 Secondary illumination via view factors. Sparse computation: only PSR↔sunlit-pixel pairs. Reflected visible + thermal IR. Single-bounce approximation. Reference implementation from Schorghofer Topo3D.

3.4 Validation. Compare illumination fraction maps against Mazarico et al. (2011). Compare PSR temperatures against Diviner Williams et al. (2019) polar products.

Chapter 4: Spatial Thermal Mapping Pipeline

4.1 Pipeline architecture. For each 20 m pixel: illumination history → 1D solver → $T(z, t)$ at 55 depth nodes → resample to 5–10 cm output grid. Parallelized with joblib/multiprocessing.

4.2 Study regions. 3–5 south pole sites (Shackleton rim, de Gerlache, Connecting Ridge, Cabeus). 5×5 km patches = 62,500 pixels each.

4.3 Results. Surface and subsurface T maps at 4 depths. Diurnal $T_{\max}$, $T_{\min}$, T_{mean} . Seasonal variation.

4.4 Diviner validation. Synthetic Diviner observations convolved to 240 m footprint. Residual maps. Target: <10 K RMS at polar latitudes.

4.5 Conductivity model comparison. Run entire pipeline with Hayne (2017) vs Martinez-Siegler (2021). Map the ΔT at each depth. Show 5–10 K differences in the top 50 cm at PSR locations.

Chapter 5: Coupling to TSUKIMI Radiative Transfer Model

5.1 Forward model. Your $T(z,t) + \rho(z)$ feeds the RTM. RTM computes synthetic $T_B(v, t)$ at TSUKIMI frequencies (~480 GHz). Incoherent multi-layer emission model with volume scattering (Wang et al. 2024).

5.2 The Jacobian coupling — joint inversion. The Jacobian matrix $\mathbf{J} = \partial T_B / \partial \mathbf{x}$ contains partial derivatives of brightness temperature with respect to regolith state vector $\mathbf{x} = [H, K_c, \rho_s, \tan\delta, \dots]$. This tells you: *which regolith parameters does the THz observation actually constrain?* A large Jacobian element means the observation is sensitive to that parameter; a small one means it cannot distinguish changes.

5.3 Iterative optimization loop. Start with thermal model guess $\rightarrow$ forward RTM $\rightarrow$ compare synthetic T_B with observed $T_B \rightarrow$ compute cost function $\chi^2 = (\mathbf{T}_{B,obs} - \mathbf{T}_{B,syn})^T \mathbf{S}^{-1} (\mathbf{T}_{B,obs} - \mathbf{T}_{B,syn}) \rightarrow$ use Jacobian to compute gradient $\rightarrow$ update regolith parameters $\rightarrow$ re-run thermal model $\rightarrow$ iterate until convergence. This is Gauss-Newton or Levenberg-Marquardt optimization.

5.4 What gets improved. Each iteration refines *both* the thermal profile and the RTM prediction simultaneously. Your thermal model provides the physics constraint (temperatures must obey the heat equation); the RTM provides the observational constraint (T_B must match what TSUKIMI measures). Together they constrain parameters that neither model could pin down alone.

5.5 Practical scope for thesis. Full implementation of the iterative loop may extend beyond September. Thesis contribution: (1) compute the Jacobian numerically (finite-difference perturbation of each parameter), (2) analyze which parameters the THz channel is most sensitive to, (3) demonstrate one iteration of the inversion on a synthetic test case, (4) design the API/data format for ongoing collaboration.

5.6 Impact of conductivity model on T_B . Run forward RTM with Hayne vs Martinez-Siegler thermal profiles. If $\Delta T_B >$ instrument noise (~0.5 K), the conductivity model choice is observable — a publishable finding with direct TSUKIMI mission relevance.

Chapter 6: Ice Survivability Mapping

6.1 Methodology. Time-averaged sublimation rates from Hertz-Knudsen equation using $T(z,t)$ output. Knudsen diffusion correction for buried ice. Stability criterion: $E < 100 \text{ kg/m}^2/\text{Gyr}$.

6.2 Ice stability depth maps. For each pixel, find shallowest depth where ice survives >1 Gyr. Surface-stable, buried-stable, and unstable classifications.

6.3 Sensitivity to conductivity model. Compare Hayne vs Martinez-Siegler ice stability maps. Quantify Δz^* (shift in stability depth) and ΔA (change in total stable area in km^2).

6.4 Connection to TSUKIMI. Overlay ice stability zones on predicted T_B anomaly maps. Identify locations where THz observations could distinguish icy from dry regolith.

Chapter 7: Conclusions and Future Work

7.1 Summary of contributions. Pipeline, sensitivity analysis, Jacobian coupling design, ice maps.

7.2 Future work. Full iterative Jacobian inversion with real TSUKIMI data. Extension to microphysical (Burger 2024) property model. Machine-learning surrogate for real-time inversion. Multi-volatile stability mapping (CO_2 , NH_3).

2. Pipeline Architecture (Data Flow)

A. DEM + Ephemeris	20 m LOLA GeoTIFF + SPICE kernels	Horizon profiles (720 az × N pixels)	You
B. Illumination	Horizons + solar ephemeris	Direct + scattered flux Q(t) per pixel	You
C. 1D Thermal Solver	Q(t), regolith properties, Q _b	T(z, t) at 55 depth nodes per pixel	You
D. Forward RTM	T(z,t), ρ(z), ε(z,ν)	Synthetic T _B (ν, t)	TSUKIMI team
E. Jacobian	Perturbation of [H, K _c , ρ _s , tanδ]	J = ∂T _B /∂x matrix	Joint
F. Inversion	J, T _{B,obs} , T _{B,syn} , priors	Updated x → re-run C→D→E	Joint
G. Ice Map	Converged T(z,t)	Ice stability depth z* per pixel	You

How the Jacobian coupling works, concretely. Think of it as two models talking to each other through derivatives.

Forward pass: Your thermal solver produces T(z,t). The RTM integrates this to predict what TSUKIMI would observe: T_B.

Jacobian computation: Perturb one parameter (e.g., increase H by 1%). Re-run thermal solver → get new T(z,t) → re-run RTM → get new T_B. The ratio ΔT_B/ΔH is one column of the Jacobian. Repeat for each parameter. With ~5 parameters, this means ~10 extra forward runs (finite-difference, central).

Update step: Using the Jacobian and the misfit between predicted and observed T_B, a Gauss-Newton step updates the parameter vector: $x_{k+1} = x_k + (J^T J + \lambda I)^{-1} J^T (T_{B,obs} - T_{B,syn})$. This is the Levenberg-Marquardt algorithm.

Your thesis scope: Compute J numerically, analyze sensitivity, demonstrate one synthetic inversion. Full operational inversion with real TSUKIMI data is future work.

3. Where Your Existing Work Lives

Nothing is thrown away. Here is exactly how each piece of your current work maps into the revised outline:

Existing asset	New role in thesis
Crank-Nicolson solver (Numba)	Core engine of Stage C. Zero changes needed.
Discrete 3-layer density model	Chapter 2.4: comparison figure showing why continuous properties are needed. This motivates the upgrade — it's pedagogically valuable.
Apollo 15/17 HFE validation	Chapter 2.7: one-paragraph sanity check proving solver correctness. Keep fig08/fig09 from your notebook.
Horizon ray-tracer (72 az, Numba)	Stage A/B. Upgrade to 720 azimuths and 20 m DEM. Core infrastructure reused.
Sky view factor computation	Chapter 3.3: basis for view-factor secondary illumination.
Solar geometry module	Chapter 3.2: upgrade to SPICE but same physics.
17 publication figures	Chapters 2 and 4: reuse DEM maps, diurnal curves, sensitivity analyses. Add new spatial map figures.
Icarus LaTeX manuscript	Refactor into thesis Chapter 2 (methods) and results sections. Pivot the standalone paper toward PSJ with spatial results.
Lunar-Clean package structure	The pipeline IS Lunar-Clean v2. Same repo, expanded scope.

4. Your Novel Contributions (What Makes This a Thesis)

#	Contribution	Why it's novel
1	Conductivity model sensitivity map for Artemis sites	No published comparison of Hayne vs Martinez-Siegler vs Burger at specific south pole locations. You quantify how model choice shifts ice stability depth by Δz and stable area by ΔA .
2	THz brightness temperature sensitivity to thermal model	Nobody has asked: does the conductivity model choice produce a distinguishable THz signal? Your unique TSUKIMI position lets you answer this.
3	Jacobian sensitivity analysis for TSUKIMI	First published Jacobian $\partial T_B / \partial x$ for THz frequencies on lunar regolith. Tells the mission which parameters it can actually constrain.
4	20 m ice survivability maps at Artemis candidate sites	Existing maps use 240 m Diviner resolution. Yours are 12× finer and include view-factor PSR treatment.
5	Open-source Lunar-Clean v2 pipeline	Community tool: DEM $\rightarrow$ illumination $\rightarrow$ thermal profile $\rightarrow$ ice map. Comparable to heat1d but with shadow modeling and spatial mapping built in.

5. Compressed Timeline (April – September 2026)

W1–2 Apr 14–Apr 27	Replace 3-layer with H-parameter + Martinez-Siegler low-T. Validate at equator + 80°S. Keep discrete model as comparison.	Updated solver with switchable property models. Comparison figure (Ch. 2.4).
W3–4 Apr 28–May 11	Download 20 m LOLA DEM. Upgrade ray-tracer to 720 azimuths. Validate illumination maps.	Illumination module on 20 m grid. Validation vs Mazarico.
W5–7 May 12–Jun 1	Implement sparse view-factor secondary illumination. Single-bounce scattered visible + thermal IR.	PSR energy budget with secondary illumination. Comparison with/without scattering.
W8–11 Jun 2–Jun 29	Build spatial pipeline driver. Run 3–5 south pole patches. Validate against Diviner polar products.	Surface + subsurface T maps at 4 depths. Diviner residual maps. (Ch. 4)
W12–13 Jun 30–Jul 13	Conductivity model comparison: run pipeline with Hayne and Martinez-Siegler. Map ΔT at each depth.	Sensitivity maps showing 5–10 K differences. (Ch. 4.5, key result)
W14–15 Jul 14–Jul 27	Forward RTM coupling. Generate synthetic T_B from thermal profiles. Compute Jacobian by finite-difference perturbation.	Jacobian matrix. Sensitivity analysis showing which parameters THz constrains. (Ch. 5)
W16–17 Jul 28–Aug 10	Ice survivability mapping. Sublimation rates from $T(z,t)$. Compare Hayne vs Martinez-Siegler ice maps.	Ice stability depth maps. Δz^* and ΔA quantification. (Ch. 6)
W18–21 Aug 11–Sep 7	Thesis writing. Manuscript pivot for PSJ submission. Code packaging for GitHub + Zenodo.	Thesis draft. PSJ manuscript. Public Lunar-Clean v2 repo.

6. This Week (April 14–20)

Monday–Tuesday: Read Martinez-Siegler (2021) Sections 2–3. Download their MATLAB code from Zenodo. Understand the low-T conductivity modification.

Wednesday–Thursday: In Lunar-Clean, create `density_hayne()` and `conductivity_martinez()` alongside your existing discrete functions. Run both at 80°S, compare subsurface T at 1 m. You should see 5–10 K difference.

Friday–Sunday: Download 20 m south pole LOLA DEM from PGDA. Load in Python with rasterio. Run your existing ray-tracer on a 100×100 pixel test patch to verify compatibility. Also download Schorghofer's Planetary-Code-Collection and read the Topo3D README.

Email Prof. Kasai: 'I am restructuring my thesis around a TSUKIMI-coupled thermal pipeline with Jacobian inversion and ice survivability mapping. The discrete model becomes a motivating comparison rather than the main result.'